

ECHS FA-06 — Beneficiary-Level Pharmacy Fraud Report

Beneficiary Pharmacy Fraud Analytics · FY 2025-26

FA-06 | Composite Risk Scoring

CLASSIFICATION	PERIOD	RECORDS ANALYSED	QUERY SETS
RESTRICTED	2025-04-01 to 2026-03-31	332,206 Beneficiaries · 895,552 Episodes	4 Fraud Types / 11,771 HIGH-risk flagged

ECHS — Center for Developing Intelligent Systems

22 May 2026

ECHS Beneficiary-Level Pharmacy Fraud Screening

FY 2025-26 · 1 April 2025 – 31 March 2026

Report date: 22 May 2026 **Methodology:** Composite risk scoring combining six normalised fraud signals weighted by hospital risk context.

Note to reviewer: This report is produced by an automated screening system. Every flagged case is a *lead for investigation*, not a confirmed finding. A qualified auditor should review each case before any action is taken.

About This Analysis

This report presents the findings of **ECHS FA-06 — Beneficiary-Level Pharmacy Fraud Analytics**, a systematic screening exercise applied to all ECCHS pharmacy claims for the financial year 2025-26.

The analysis assigns each beneficiary a composite fraud score (0–1) by combining six independent normalised signals into a weighted sum. The primary discriminator is the **hospital context weight (35%)**: a beneficiary whose pharmacy claims are concentrated at hospitals already flagged for pharmacy abuse is categorically more suspicious than one with identical episode counts at low-risk hospitals.

Four distinct fraud types are detected:

- **T1 Over-Prescribing** — Excessive OPD pharmacy episode frequency at a single facility
- **T2 IPD Inflation** — Inpatient claims where pharmacy exceeds 50% of gross billing
- **T3 Collusion** — All claims concentrated at ≤ 2 HIGH-tier hospitals
- **T4 Multi-Hospital Shopping** — Pharmacy split across ≥ 3 hospitals

Data Overview

Metric	Count	Notes
Total beneficiaries analysed	332,206	Distinct ECCHS card holders with pharmacy claims in scope
Total pharmacy episodes (claims)	895,552	FY 2025-26, all admitted beneficiaries
Unique hospitals in scope	11,859	Hospitals with ≥ 5 claims
HIGH-tier hospitals (pharmacy > 90%)	197	Primary fraud context
Beneficiaries flagged HIGH risk	11,771	Score ≥ 0.65
Beneficiaries flagged MEDIUM risk	205,125	Score ≥ 0.4
CRITICAL-severity individual claims	2,162	IPD high-pharmacy + large claim

1. Executive Summary

A total of **332,206 beneficiaries** with pharmacy claims were analysed for the period 2025-04-01 to 2026-03-31. The analysis combined six fraud signals into a composite score (0–1), weighted by the risk profile of hospitals visited by each beneficiary.

Risk Tier	Beneficiaries	Pharmacy Billed	% of Total Billed
HIGH (score ≥ 0.65)	11,771	Rs 7,879.3L	20.8%
MEDIUM (score ≥ 0.4)	205,125	Rs 25,081.0L	66.2%
LOW	115,310	—	—
Total	332,206	Rs 37,914.6L	100%

Key finding: 11,771 beneficiaries (3.5% of all) are flagged HIGH risk, accounting for Rs 7,879.3L (20.8% of total pharmacy billing). The dominant pattern is **T3:Collusion** (10,973 cases, 93% of HIGH risk) — beneficiaries whose pharmacy claims are exclusively concentrated at hospitals already flagged for pharmacy abuse.

2. Scoring Methodology

The composite fraud score is a weighted sum of six normalised signals (each scaled 0–1):

Signal	Weight	Rationale
Hospital Context	35%	Amount-weighted avg risk tier of visited hospitals. A beneficiary at a HIGH-tier hospital is categorically more suspicious.
Total Annual Spend	15%	Log-normalised. Unusually high pharmacy spend relative to peers.
IPD Ratio	15%	Fraction of episodes that are inpatient. IPD claims should not be pharmacy-dominant.
Max Single Claim	15%	Log-normalised single-claim spike above 99th percentile (Rs 39,476).
Episode Frequency	12%	Number of distinct pharmacy visits per year.
Hospital Diversity	8%	Multi-hospital pharmacy shopping behaviour.

Hospital tier weights used for the context signal:

Tier	Overall pharmacy %	Risk weight
HIGH	> 90%	1.00
MEDIUM	> 60%	0.70
LOW	> 30%	0.40
Normal	$\leq 30\%$	0.10
Pure (standalone pharmacy)	= 100%	0.05 (expected)

The IPD hospital tier receives a **1.25× boost**: pharmacy domination within an inpatient claim is a stronger indicator of manipulation than in an OPD context.

3. Hospital Risk Context

Of 11,859 hospitals analysed:

Hospital Tier	Count	Pharmacy Billed
HIGH	197	Rs 6,592.2L
MEDIUM	418	Rs 14,014.1L
LOW	327	Rs 9,102.3L
Normal	181	Rs 1,395.5L
Pure	10,676	Rs 4,080.6L

197 HIGH-tier and **418 MEDIUM-tier** hospitals form the primary risk context. Beneficiaries who predominantly visit these hospitals receive elevated hospital context scores regardless of their own episode frequency.

4. Fraud Pattern Analysis

4.1 Fraud Types (HIGH-Risk Beneficiaries)

Fraud Type	Description	Beneficiary count	% of HIGH
T3:Collusion	All claims at HIGH-tier hospitals (≤ 2 hospitals)	10,973	93.2%
T2:IPD-Inflation	IPD claims dominated by pharmacy ($> 50\%$)	1,816	15.4%
T1:Over-Prescribing	Excessive OPD pharmacy episodes	478	4.1%
T4:Multi-Hosp	Pharmacy split across ≥ 3 hospitals	122	1.0%

4.2 Pattern Observations

- **T3 Collusion (10,973 cases, 93%):** The dominant pattern. These beneficiaries channel nearly all pharmacy claims through ≤ 2 hospitals that are themselves flagged as HIGH/MEDIUM-tier pharmacy abusers. This concentration is consistent with fictitious prescriptions, duplicated billing, or coordinated kickback arrangements between the beneficiary and a specific hospital.
- **T2 IPD Inflation (1,816 cases, 15%):** 608 beneficiaries have at least one inpatient claim where pharmacy exceeded 50% of gross billing. IPD admissions should include procedures, room charges, OT, and diagnostics — pharmacy domination is anomalous and suggests inflated or fictitious pharmacy line items within inpatient claims.
- **T4 Multi-Hospital Shopping (122 cases):** Beneficiaries visiting ≥ 3 hospitals for pharmacy. This may indicate "prescription shopping" — obtaining multiple prescriptions from different facilities to maximise the total claimable amount.
- **T1 Over-Prescribing (478 cases):** High OPD pharmacy episode frequency without the above patterns. May indicate genuine chronic illness or systematic over-prescription at a single facility.

5. Claim-Level Flags

Total flagged claims (HIGH + MEDIUM beneficiaries): **684,536**

Flag	Description	Count
<code>flag_ipd_high_pharmacy</code>	IPD claim with pharmacy $> 50\%$ of gross billing	3,373
<code>flag_large_claim</code>	Single claim above 99th pct ($> \text{Rs } 39,476$)	8,601
<code>flag_high_hosp</code>	Claim at a HIGH or MEDIUM tier hospital	574,073

Flag	Description	Count
flag_ipd_episode	Inpatient (IPD) episode	19,662

2,162 claims carry CRITICAL severity (simultaneously IPD high-pharmacy + large claim) — these represent the highest-priority individual claims for investigation.

6. Top 25 HIGH-Risk Beneficiaries

#	Benf ID	Name	Episodes	Spend (Rs)	Score	Fraud Pattern	
1	3462368	RAJBIR KAUR	7	Rs 2,424,954	0.9299	T2:IPD-Inflation	T3:Collusion
2	2141915	RAJBAHADUR UPADHYAY	14	Rs 2,903,829	0.9089	T2:IPD-Inflation	T3:Collusion
3	2127629	OMEY LAL	9	Rs 422,844	0.9067	T2:IPD-Inflation	T3:Collusion
4	4197225	VIJAISHWAR PRATAP SINGH	16	Rs 8,015,914	0.8998	T2:IPD-Inflation	T3:Collusion
5	2880922	ANOUSHKA MISHRA	17	Rs 1,879,135	0.8993	T2:IPD-Inflation	T4:Multi-Hosp
6	1650113	INDRAJIT SINGH	27	Rs 2,397,304	0.8919	T2:IPD-Inflation	
7	2443042	RAMCHANDRA MALUSARE	6	Rs 132,267	0.8890	T2:IPD-Inflation	T3:Collusion
8	4095326	OM PRAKASH SINGH	12	Rs 174,798	0.8884	T2:IPD-Inflation	T3:Collusion
9	4036289	PALWE DWARKA VITTHAL	15	Rs 1,753,078	0.8729	T2:IPD-Inflation	
10	2088750	KARUNA BHANDARI	7	Rs 359,819	0.8697	T2:IPD-Inflation	T3:Collusion
11	2925128	SHUBHAM CHOUDHARY	5	Rs 483,235	0.8577	T2:IPD-Inflation	T3:Collusion
12	2867011	KRISHAN KUMAR	15	Rs 5,627,459	0.8562	T2:IPD-Inflation	T4:Multi-Hosp
13	3364375	RAM PRATAP SINGH	16	Rs 351,730	0.8492	T3:Collusion	
14	2660040	AASIYA AKHTAR	2	Rs 683,155	0.8484	T2:IPD-Inflation	T3:Collusion
15	3364361	SUMAN DEVI	14	Rs 334,526	0.8482	T3:Collusion	
16	2277706	VIJAYKUMAR BALIRAM YAD	2	Rs 231,172	0.8473	T2:IPD-Inflation	T3:Collusion
17	1930806	IRA SHARMA	11	Rs 174,771	0.8462	T3:Collusion	

#	Benf ID	Name	Episodes	Spend (Rs)	Score	Fraud Pattern	
18	2943663	PATIL SUMAN SUBHASH	12	Rs 815,695	0.8410	T2:IPD-Inflation	T4:Multi-Hosp
19	1969934	SHIRISH JAYRAM SALUNKH	5	Rs 274,471	0.8401	T2:IPD-Inflation	T3:Collusion
20	3186855	HARGOPAL SINGH	4	Rs 230,955	0.8400	T2:IPD-Inflation	T3:Collusion
21	1724651	MANOJ MANGAL	15	Rs 364,099	0.8389	T3:Collusion	
22	3090674	SUDHIR KUMAR	11	Rs 265,229	0.8389	T3:Collusion	
23	1506585	NAKUL KUMAR SAHOO	10	Rs 146,646	0.8376	T3:Collusion	
24	2862019	RASMITA PANIGRAHI	10	Rs 131,029	0.8373	T3:Collusion	
25	3039555	SERBATHY	9	Rs 413,815	0.8367	T3:Collusion	

Generated by ECHS FA-06 automated screening system — 22-May-2026